DSA 210 Project Proposal

Project Title:
The Impact of Travel Distance and Weather Conditions on UEFA Champions League Away Performances

I will use the publicly available **Football Data from Transfermarkt** dataset from Kaggle as my primary data source. More specifically, I will work with the **games.csv** file, which contains historical match-level records from European football competitions. Since the project guidelines require enriching a public dataset with additional data, I plan to combine this dataset with two external sources: a geocoding step for obtaining location coordinates and a historical weather API for match-day weather information.

The main dataset will be downloaded from Kaggle in CSV format and processed in Python using the Pandas library. I will filter the data to keep only **UEFA Champions League** matches by using the relevant competition identifier. For the enrichment stage, I will map team or stadium locations to geographic coordinates through a geocoding process, and then calculate the approximate geodesic travel distance in kilometers between the away team's location and the match location. As a second enrichment, I plan to use a free historical weather API, such as **Open-Meteo**, to retrieve match-day weather variables for the host location based on the match date and geographic coordinates. These external data sources will be merged with the original match records to form the final analysis dataset.

The resulting dataset will be a structured tabular dataset consisting of UEFA Champions League matches across multiple seasons. Each row will represent a single match. The original variables will include categorical and numerical features such as the home team, away team, date, and match outcome. After enrichment, the dataset will also include newly engineered numerical variables such as **travel distance (km)** and selected **match-day weather measures**. These enriched features will allow me to investigate whether physical and environmental factors are associated with away-team performance, and they will also serve as inputs for later statistical analysis and machine learning models.